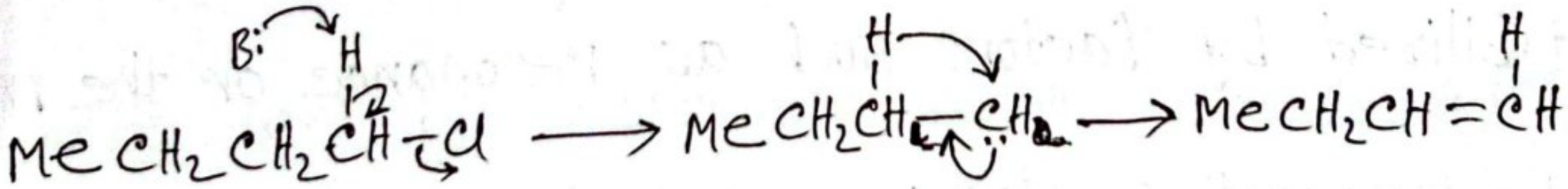
2017(t)

What do you meant by 1,1-elimination reaction. Give example?

⇒ 1,1-elimination reaction is a type of chemical reaction where two substituents are removed from the same atom, resulting in the formation of a double bond. This reaction typically leads to the formation of an alkene. This is a relatively rare type of elimination reaction, as most elimination reactions are 1,2-elimination, where the substituents are removed from adjacent carbon atom. This 1,2-elimination is often called a "beta-elimination". 1,1-elimination is called α-elimination. Example:



2016, 2021, 2015

Write short note on E_1cB with mechanism & example?

⇒ the E1cB (Elimination, unimolecular conjugate base) mechanism is a type of elimination reaction in organic chemistry. It involves a two-step process

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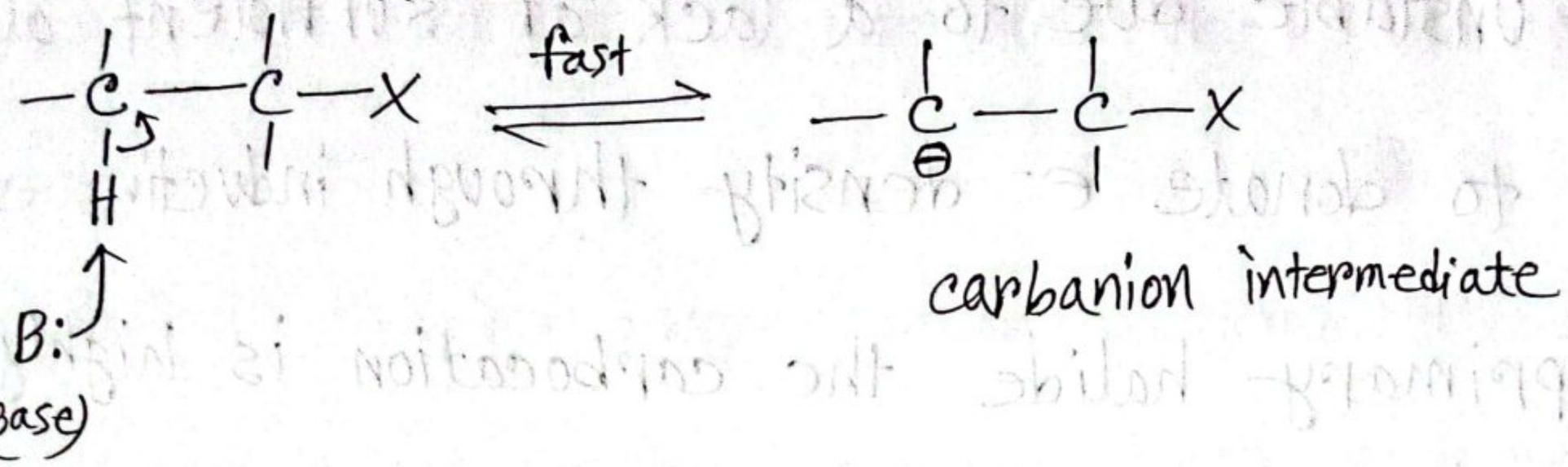
Where the first step is the formation of a carbanion (a carbon with a negative charge) by removing a proton (H^+) from a carbon adjacent to a leaving group. Secon. Step is the elimination of the leaving group (such as a halide) to form a double bond. This mechanism usually occurs when the substrate has a relatively acidic H atom adjacent to the leaving group & when a strong base is used.

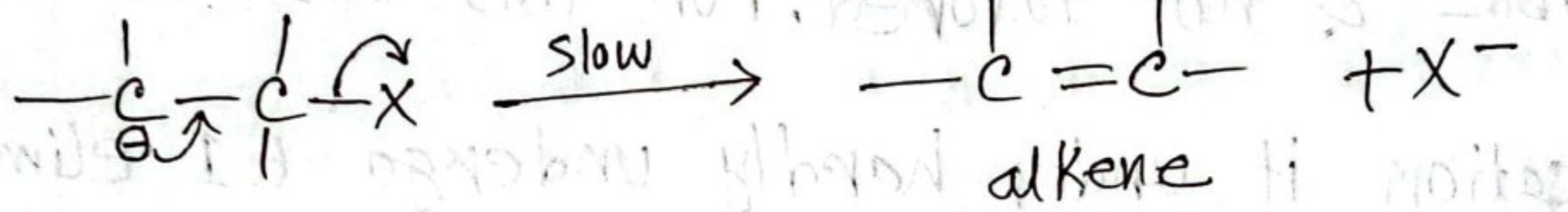
Mechanism: ⓐ A. Strong base removes a proton from the carbon adjacent to the leaving group, forming a carbanion. This step is usually reversible. This carbanion is stabilized by factors such as resonance or the presen of electronegative atoms nearby.

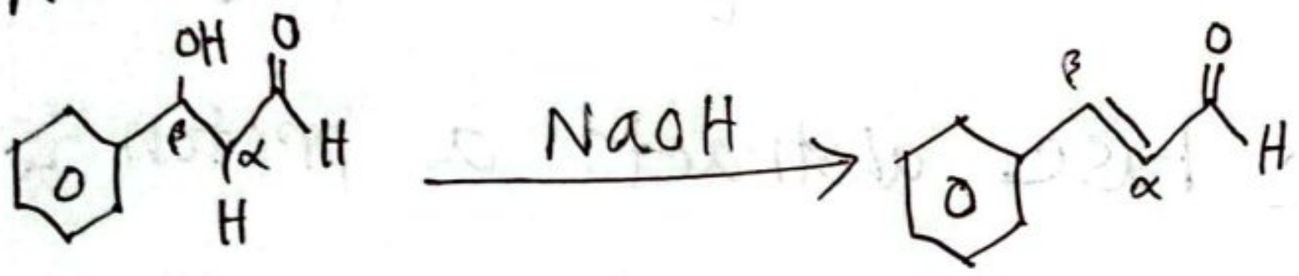
ⓑ The carbanion then expels the leaving group to form a double bond & producing an alkene. This step is irreversible.

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step-1: 

step-2: 

example: 

2018, 2016
(X) primary halides hardly undergo E1 elimination? explain?

⇒ primary halides hardly undergo E1 elimination because the E1 mechanism involves the formation of a carbocation intermediate: In a primary halide, the carbon bearing the leaving group is a primary carbon, meaning it is only attached to one other carbon atom. This makes the resulting carbocation

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highly unstable due to a lack of sufficient alkyl groups to donate e⁻ density through inductive effect. For primary halide the carbocation is highly unstable & not favored. For this unstable nature of carbocation it ~~under~~ hardly undergo E1 elimination. primary halides are more likely to undergo E2 elimation, where the base abstracts a proton & a leaving group in one step. If strong base is present, E2 elimination can occur.

2018.
⊗ Write effect of e⁻ withdrawing groups in the elimination reaction?

⇒ e⁻ withdrawing group (EWG) withdraw e⁻ density from the carbon atom to which they are attached. This makes the adjacent carbon (β carbon) more e⁻ deficient & there H more acidic.

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(II) In elimination reaction, EWG makes the proton removal easier & faster.

(III) EWG pull e density away from the carbon atom bearing the leaving group (LG). This weakens the bond btwn the carbon & the LG, making the LG more prone to depart.

(IV) In E1cB, when strong EWG are present, the formation of a carbanion intermediate becomes more favorable.

(V) EWG favor the E1cB mechanism over E1 & E2.

(VI) EWG can also influence the ragioselectivity of elimination reaction

(VII) In E1, EWG can stabilize the intermediate carbocation by delocalizing the positive charge. EWG increase the rate of E1 reaction by stabilizing the carbocation.

(VIII) In E2 reaction, EWG can stabilize the transition state. & Enhance the reaction rate by making H more acidic.

Such EWG are $-NO_2$, $-CN$, $>C=O$, $-CHO$ etc.

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(A) 2017 Elimination is promoted by e⁻ withdrawing substitution. Explain with example?

⇒ This statement refers to the effect of EWG on elimination reaction. In E1 reaction, the EWG helps to stabilize the carbocation through inductive effect & promoting the elimination.

In E2 reaction, the presence of EWG near the leaving groups helps to stabilize the ~~carbanion~~ negative charge that develops on the adjacent carbon during the transition state & promotes the elimination reaction.

EWG pull e⁻ away from the carbon they are attached to. By pulling e⁻ away from the carbon EWG developing negative charge, EWG help stabilize the transition state this makes the elimination reaction more favorable.

EWG stabilize the carbanion intermediate formed in the first step of E1cB reactions & promoting E1cB reaction

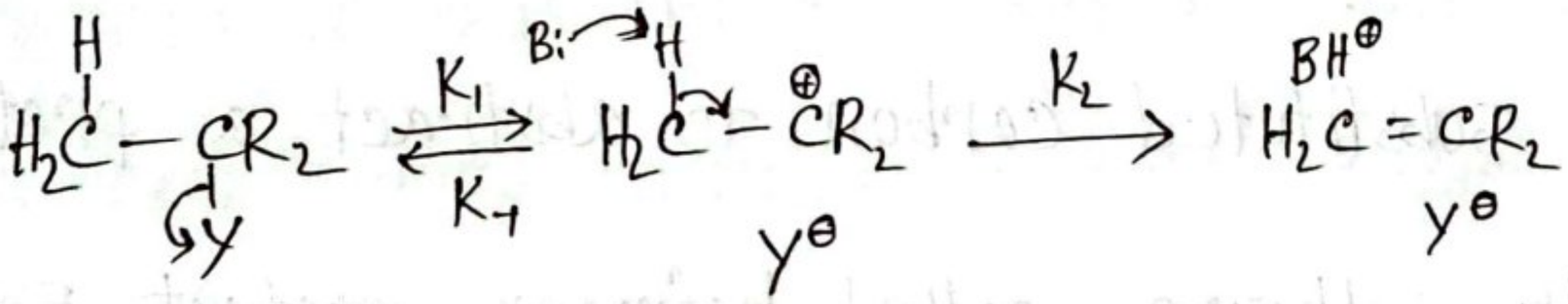
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2017, 2017 (t)

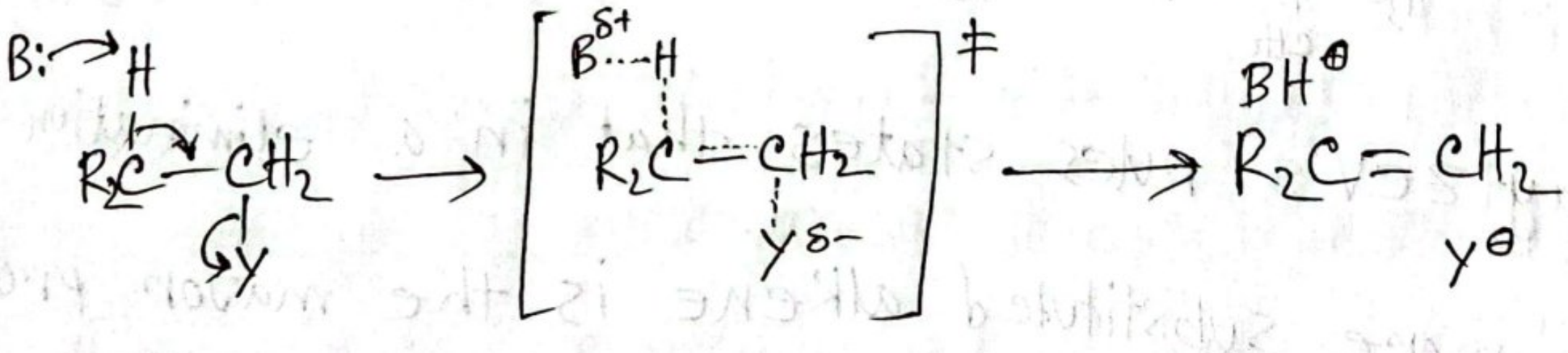
⊗ show three dfrnt schematic representation for 1,2-elimination?

⇒ Three different schematic representations of the 1,2-elimination : E1, E2 & E1cB. these 1,2-elimination reaction involves the removal of two substituents (such as proton & leaving group) from adjacent carbon atoms, resulting in the formation of a double bond.

If the C–Y bond ~~is broken first~~ & H–C bonds can be broken seperately in two-step process. If the C–Y is broken first & carbocation is formed, this is E1.



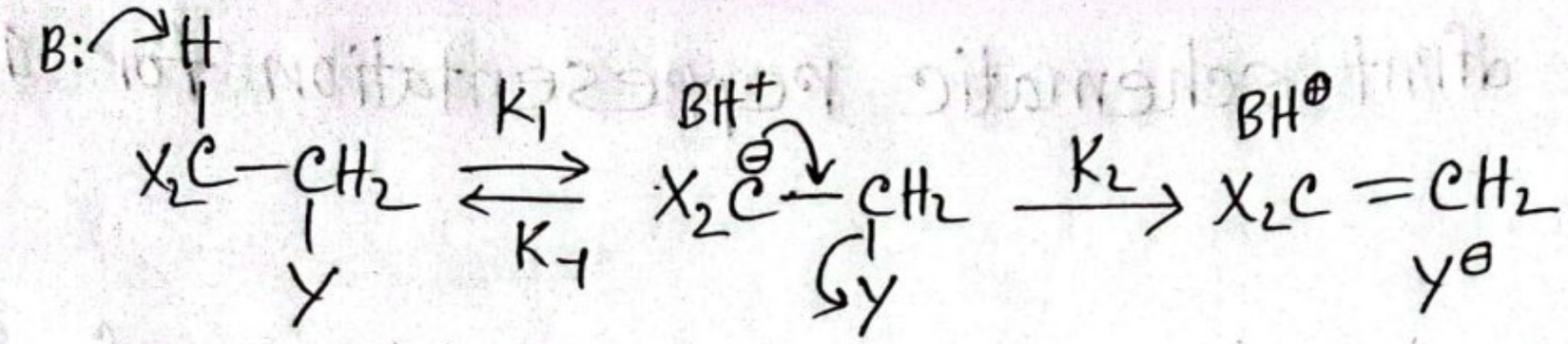
If H–C & C–Y breaking in one step process, passing through a single T.S. This is E2 mechanism.



If the H–C bond broken first & involving a carbanion intermediate, this mechanism is E1cB.

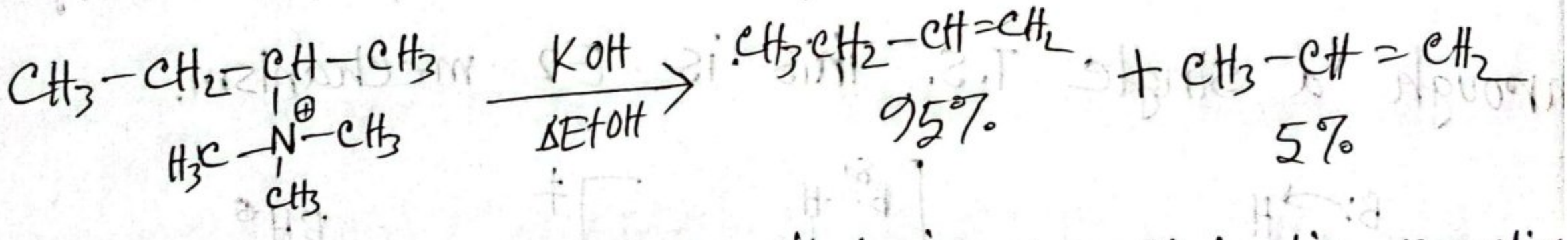
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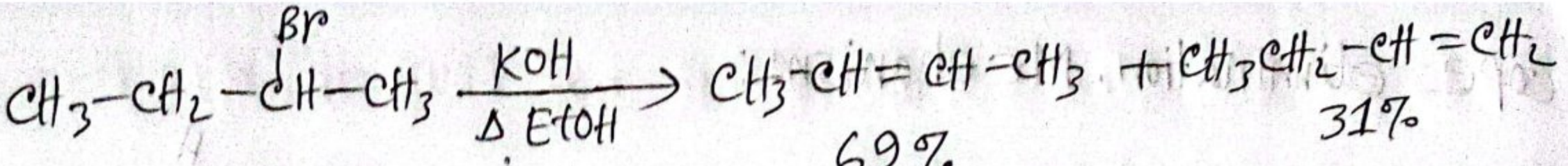
(2016)
what are Hofman & saytzev rules of elimination? Explain the rules using appropriate examples?

⇒ Hofmann elimination is an elimination reaction of an amine to form amines. It state's that if you use a bulky (hindered) base in an elimination reaction, the less substituted alkene will be the major product. This is because the bulky base has difficulty accessing the more substituted carbon to abstract a proton. And least stable alkene called hofmann product. Example



⇒ Saytzev's rules states that in a elimination reaction the more substituted alkene is the major product. because the more substituted alkene is generally more stable. Example:

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Hofmann's rule is the opposite of saytzeff's rule. It applies when there is significant steric hindrance, especially with large leaving groups like quarternary ammonium salts. generally saytzeff's rule is applies when there is no steric hindrance & Hofmann's rule is often observed when bulky groups are involved, such as in reactions with quarternary ammonium salts.

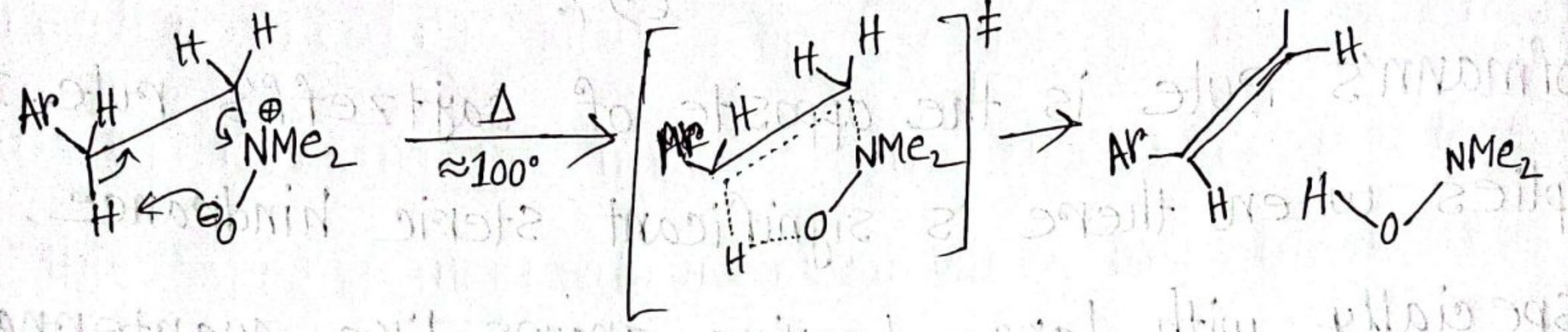
2016, 2017(t)

Ⓧ Give the mechanism of pyrolytic SYN elimination reaction considering tertiary amine oxide & xanthates as examples?

⇒ Both tertiary amine oxides and xanthates undergo pyrolytic syn elimination reactions, which involve the elimination of groups from the same side of the molecule through a mechanism. These reactions proceed through cyclic transition states and forming alkenes. pyrolytic syn elimination in tertiary amine oxides is called

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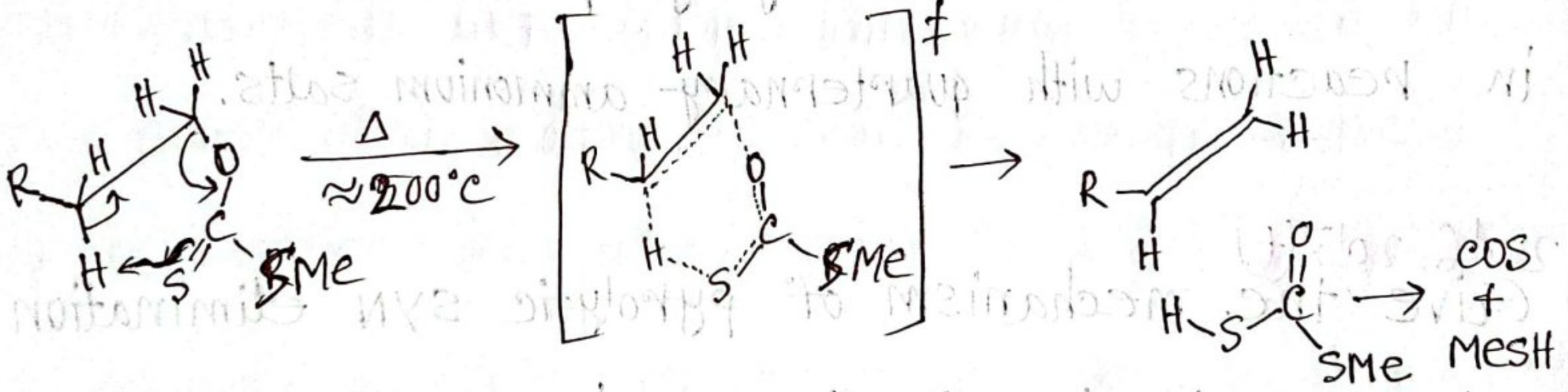
Cope elimination. Mechanism:



The pyrolysis of xanthates is called chugaev reaction.

In the transition state, six-membered ring is formed.

Mechanism:



pyrolytic syn elimination also known as Ei eliminations (elimination intra-molecular).

2017(t)
⊛ under comparable conditions, CH_3CH_2Br yielded ≈1% alkene while $PhCH_2CH_2Br$ yielded ≈99% alkene - explain?

⇒ the primary carbocation formed from CH_3CH_2Br is very unstable. Hence, the elimination is not favoured

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and very little alkene is produced which is 1%. But the benzylic carbocation formed from $PhCH_2CH_2Br$ is very stable due to resonance with the phenyl ring. The positive charge can be delocalised onto the phenyl ring, making the carbocation much more stable than a simple primary carbocation. The greater stability of the benzylic carbocation makes the elimination reaction much more favourable, resulting in 99% alkene production.

2016
⊗ What are the possible grounds which can be stated for favouring E2 elimination of acyclic molecules from the anti-periplanar conformation?

⇒ The E2 elimination mechanism involves the simultaneous removal of a proton (H^+) from a β-carbon & the departure of a leaving group from the α-carbon. This process is highly depend on the geometry of the molecule, specifically the antiperiplanar conformation, which is the most favourable for E2 elimination.

① In an antiperiplanar arrangement, the C-H and C-LG

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bonds are in the same plane but point in opposite directions; creating ideal geometry for interaction of the orbitals, and making E2 elimination more favourable.

2. This arrangement allows maximum overlap btwn the orbitals of the leaving group, the hydrogen, and the developing π system of the forming double bond in the alkene.

3. The antiperiplanar conformation facilitates a smooth, low-energy transition state, leading to a faster r. rate.

4. Antiperiplanar arrangement has a lower activation energy for the reaction compared to other conformation.

5. As the C-H and C-LG bonds aligned in antiperiplanar orientation so deprotonation and the departure of the leaving group can occur simultaneously.

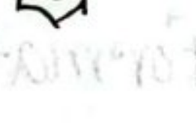
6. This conformation reduces steric hinderance which allows the base to more easily abstract the β-hydrogen, making the reaction more efficient.

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2021
(iv) Write limitations of the reaction (wolf kishner) with example?

⇒ (i) This reaction is unsuitable for substrates with base sensitive groups (epoxides or halides). Ex. $ClCH_2COCH_3$, the chloro group may undergo elimination or substitution under strong basic conditions.

(ii) this reaction uses polar solvents like ethylene glycol, limiting its application for substrates insoluble in these solvents. Ex. Highly non-polar ~~sol~~ Ketones like  may have poor solubility & slower reaction rates.

(iii) Initial step involves the formation of hydrazone intermediate, which can be challenging with sterically hindered carbonyl compounds.

(iv) It requires high temp. & strong basic conditions, which can limit its applicability to heat-sensitive substrates.

2022, 2020, 2016, 2015, TU
(x) How does trapping of intermediate help to investigate r. mechanism?

⇒ Trapping of an intermediates is a technique that can help to investigate a reaction mechanism by allowing the identification of short lived & unstable intermediates.

During a reaction, a short-lived intermediate species forms, then trapping agent added to the reaction mixture. Trapping agent

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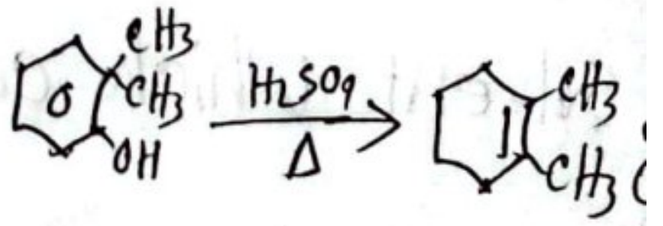
react with intermediate. the trapped product is isolated & analyzed using spectroscopic technique (IR, NMR, mass spectroscopy) to determine its structure & properties. EX. Benzyne intermediate.

Benzyne is highly reactive species that can be generated through various methods, including the elimination of a halide ion from an aryl halide. To trap benzyne, a diene such as furan can be added to the reaction mixture. The diene undergoes a diels-alder reaction with benzyne, forming an adduct. By isolating this adduct & analyzing it we gain information about its reactivity & geometry. Trapping of intermediates proposed mechanisms by providing experimental evidence.

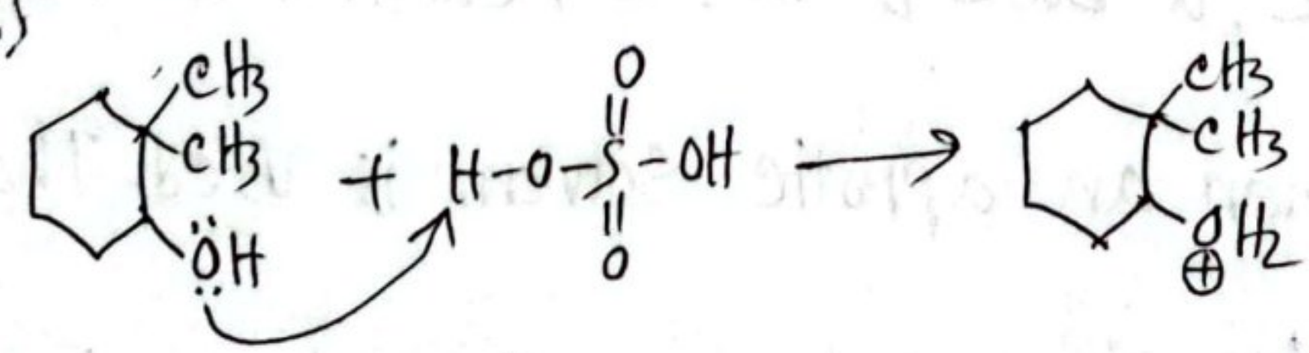


2017, 2015

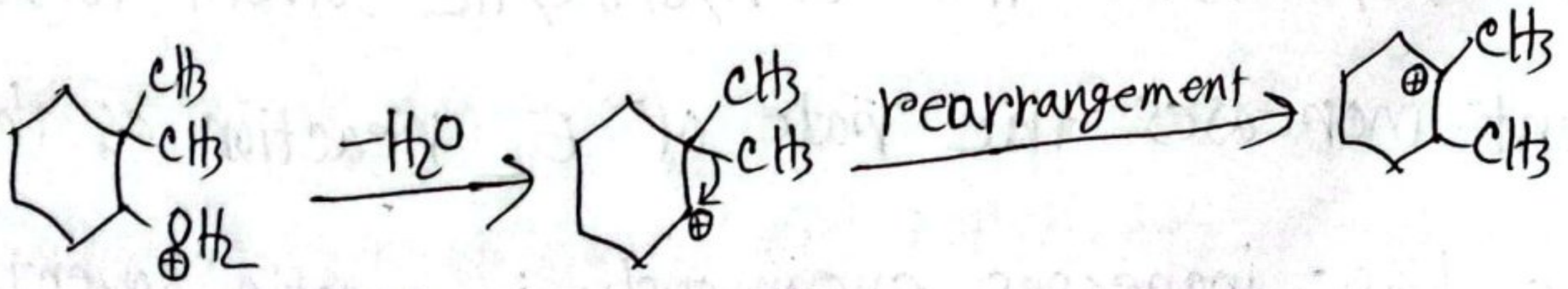
(X) Propose a detailed, stepwise mechanism for:



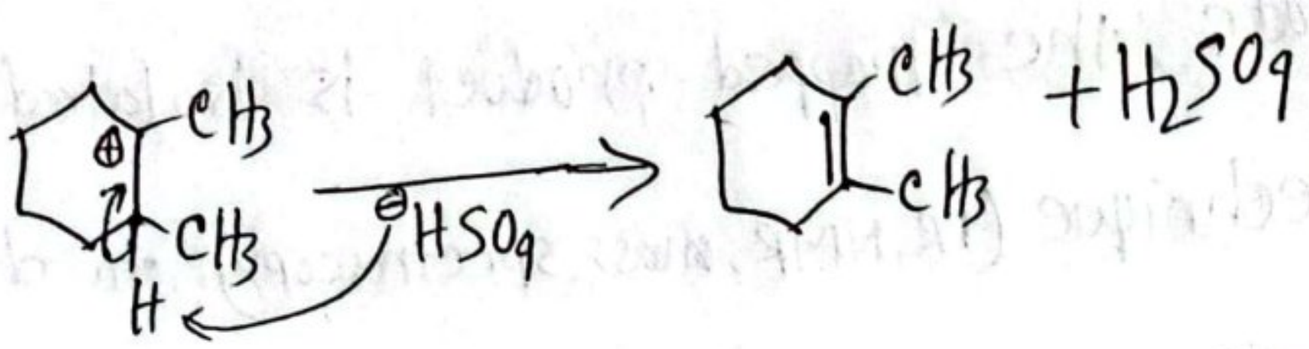
⇒ step-1:



Step-2:



Step-3:



2017
(*) with a given base, transfer from a hydroxylic solven like H_2O/EtOH to a polar aprotic one like DMF/DMSO can have Profound effect on the rate of E_2 reactions - explain?

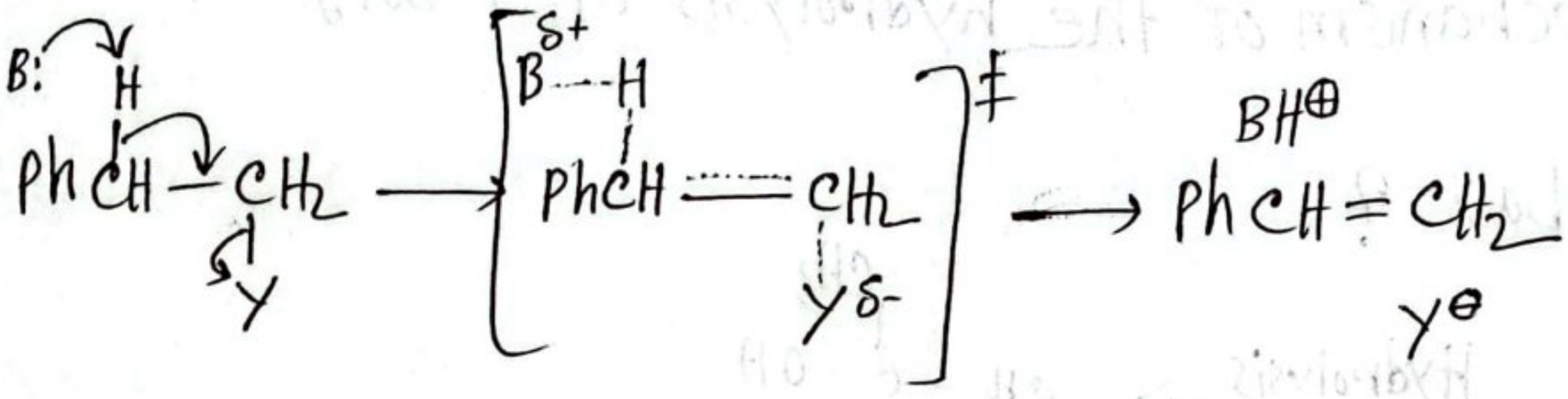
⇒ In E_2 reaction, as the strength of the base increases, the rate of reaction increases too. In polar protic solvents or hydroxylic solvents such as H_2O/EtOH, the base is solvated in the solvent due to H-bonding. This solvation hinders the base's ability to deprotonated a substrate effectively. But in polar aprotic solvents, the base do not form H-bond & do not get solvated. Hence, the base acts as a stronger base in aprotic solvent which can ~~act~~ now abstract a proton from the substract easily. Therefore, a base is more reactive towards the substrate for E_2 reaction when an aprotic solvent is used. That is, with a given base, transfer from a hydroxylic solvent to an aprotic solvent increases the rate of E_2 reaction as the strength of the base increases enormously in aprotic solvents.

2017
⊛ Explain the relative rate sequence observed for $PhCH_2CH_2Hal$ with $\bar{O}Et/EtOH$:

	$PhCH_2CH_2F$	$PhCH_2CH_2Cl$	$PhCH_2CH_2Br$	$PhCH_2CH_2I$
R. rate	1	70	4.2×10^3	2.7×10^9

?

⇒ From the Mechanism of E_2 elimination reaction we see:



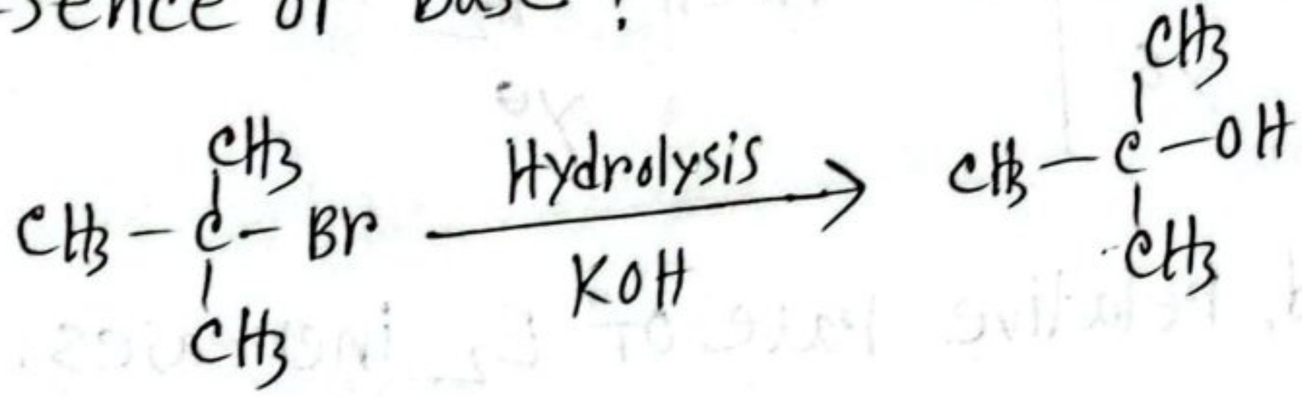
the weaker C-Y bond, relative rate of E_2 increases. As the size of halogen I > Br > Cl > F so, bond strength sequence is C-F > C-Cl > C-Br > C-I So the leaving group F is good than other. So the R. rate among them like above.

2017
⊛ which reagent in each of the following pairs is more nucleophile & why: ① $(CH_3)_3B$ & $(CH_3)_3N$ ② H_2O & H_2S ?

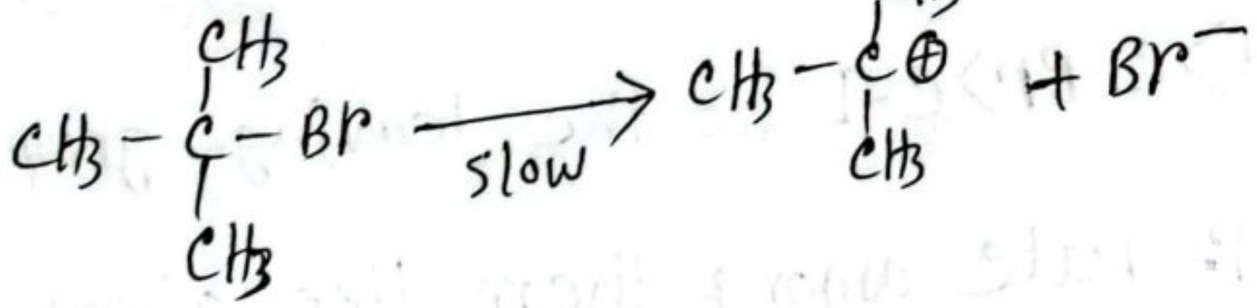
⇒ ① $(CH_3)_3N$: is more nucleophile than $(CH_3)_3B$. because N atom has a lon pain & methyl group donate e^- to the N atom so it can attack any positive charge part easily. But B has incomplete octate, it ~~contain only six elect~~ has lacks of lone pair. It has an empty p-orbital, making it e^--deficient. it tends to accept e^- rather than donate ~~e~~ e^-, so $(CH_3)_3N$ is more nucleophile.

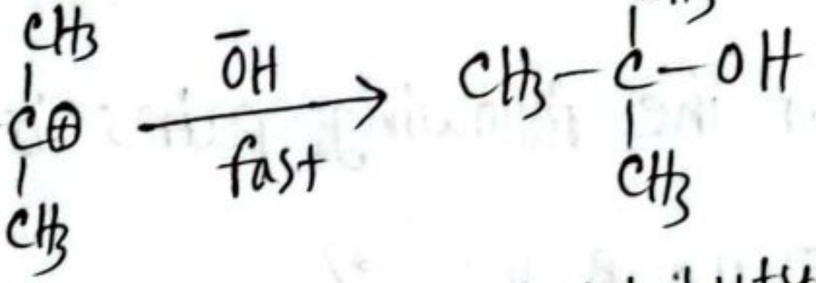
(11) $H_2\ddot{S}:$ is more ~~electropl~~ nucleophile than $H_2\ddot{O}:$. Both O & S has two lone pair but O is more electronegative than S. So O can't easily donate its lone pair than S. So H_2S more nucleophile.

(X) 2017 Give the mechanism of the hydrolysis of t-butyl bromide in presence of base?

$\Rightarrow$ 

t-butyl bromide

Mechanism: step-1: 

step-2: 

tert-butyl alcohol

It follow S_N1 mechanism & occur in 2 step.

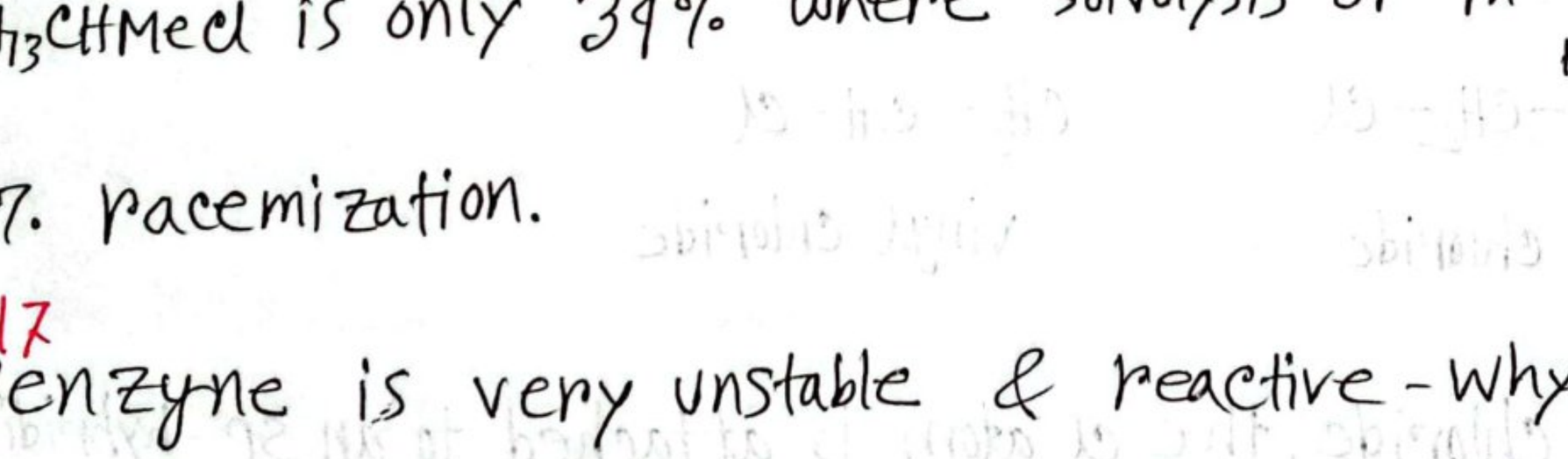
(X) 2017 Solvolysis of (+)PhCHMeCl leads to 98% racemization in aqueous acetone while (+)-C_6H_{13}CHMeCl leads to only 34% racemization-why?

$\Rightarrow$ we know, when the carbocation is more stable, then the racemization is more.

(+)-phCHMeCl $\xrightarrow[\text{slow}]{-Cl^-}$ Ph—C(Me)(H)⊕ + Cl^-

(+)-C_6H_{13}CHMeCl $\xrightarrow[\text{slow}]{-Cl}$ C_6H_{13}—C(Me)(H)⊕ + Cl^-

among these carbocation Ph—C(Me)(H)⊕ is more stable due to presence of benzene ring. For benzene ring there delocalization of e^- or resonance is possible for this reason it is stable.



but, C_6H_{13}—C(Me)(H)⊕ is less stable than Ph—C(Me)(H)⊕ so solvolysis of (+)C_6H_{13}CHMeCl is only 34% where solvolysis of Ph—C(Me)(H)⊕ leads to 98% racemization.

2017
Benzyne is very unstable & reactive - Why?

⇒ due to its highly strained triple bond in a cyclic compound. Benzyne is typically formed by removing two H or substituting groups from adjacent C-atom in benzene, creating a triple bond. This triple bond introduces significant strain & disrupts the aromatic stability of benzene means it loses aromaticy.

The triple bond in benzyne is not a typical linear triple bond. It is a bent bond, meaning the two π orbitals involved in the bond are not perfectly aligned. This misalignment creates significant angle strain, making the molecule unstable. This strain results in a high energy state for benzyne, making it highly reactive. The π-e^- in the triple bond are highly localized making benzynes electrophilic & reactive toward nucleophiles. Nucleophiles readily attack benzynes at the strained triple bond.



Benzyne

2017

Ⓐ compared to ~~alkyl~~ allyl chloride, vinyl chloride is very unreactive towards nucleophilic substitution? Why?

⇒ $CH_2=CH-CH_2-Cl$ allyl chloride

$CH_2=CH-Cl$ vinyl chloride

① In vinyl chloride, the Cl atom is attached to an sp^2-hybridized carbon atom in a double bond system. This bond has more s-character making it shorter, stronger & less polar, which reduces the tendency of Cl to leave as a leaving group.

In allyl chloride, the Cl atom is attached to an sp^3 hybridized

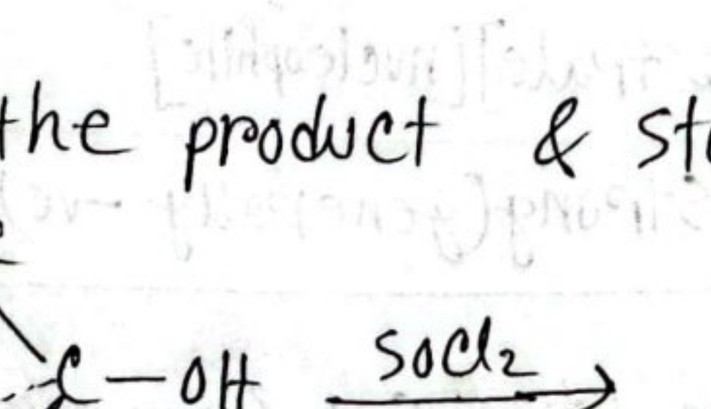
carbon, making the bond weaker & more prone to cleavage.

(II) In vinyl chloride, resonance delocalization occurs btwn the Cl lone pairs & the π-systems of the double bond. This gives the C-Cl bond partial double bond character, further stabilizing the bond & making it less reactive. But in Allyl chloride no delocalization occur.

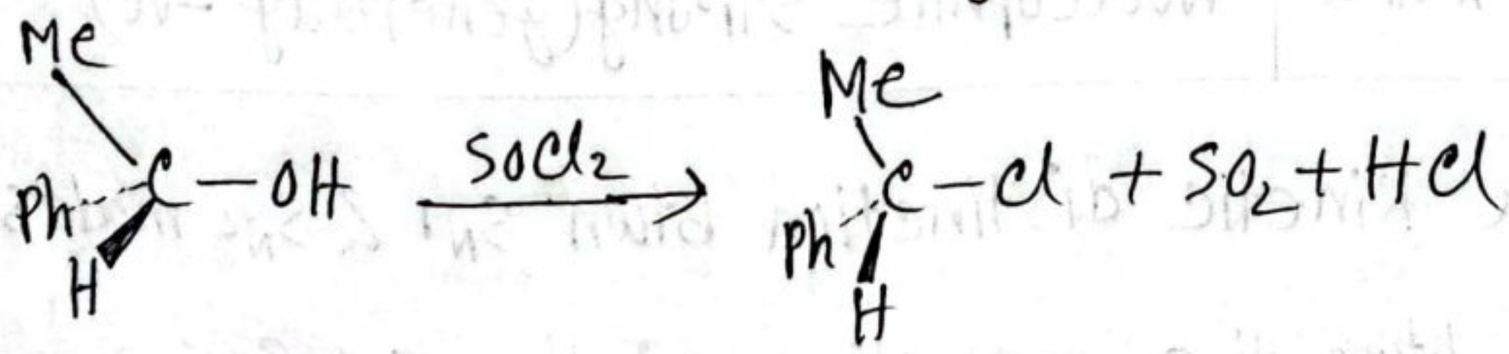
(III) The allyl carbocation is resonance-stabilized, making the reaction more favorable. Vinyl less reactive due to lack of resonance stabilization.

(IV) $CH_2=CH-CH_2-Cl \xrightarrow[\text{slow}]{-Cl^-} CH_2=\overset{\curvearrowright}{CH}-\overset{\oplus}{C}H_2 \longleftrightarrow \overset{\oplus}{C}H_2-CH=CH_2$

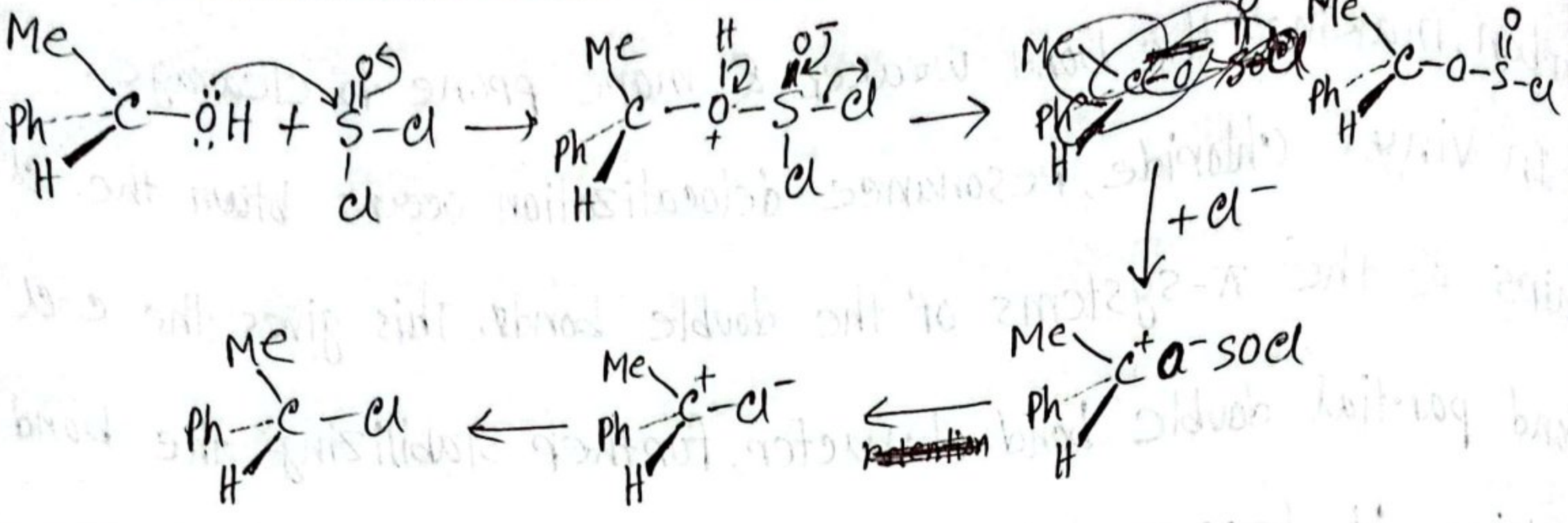
(X) 2017 what is S_Ni ? complete reaction & show its mechanism:



⇒ S_Ni means Nucleophilic Substitution intra-molecular. & in this reaction retention of configuration occurs that means configuration of the product & starting material is same.



Mechanism:



2016, 2015

⊛ What is the salient difference btwn S_N1 and S_N2 pathway?

S_N1	S_N2
two step reaction	one step reaction
carbocation formed	no carbocation formed
intermediate formed (carbocation)	~~intermediate~~ formed transition state
polar protic solvent is used	polar aprotic solvent is used
It is unimolecular reaction	bimolecular
follow first order kinetics	follow second order kinetics
partially racemized product is formed + inverted (mix)	only inverted product is formed.
Rx reactivity order $3^\circ > 2^\circ > 1^\circ > CH_3X$	$CH_3X > 1^\circ > 2^\circ > 3^\circ$ ← reactivity order
Rate = K[substrate]	Rate = K[substrate][nucleophile]
Nucleophile weak (generally neutral	Nucleophile strong (generally -ve)

2016, 2015

⊛ How doo you make a kinetic distinction btwn S_N1 & S_N2 modes?

⇒ A kinetic distinction btwn the operation of the S_N1 & S_N2 modes can be made by adding a competting nucleophile, such as azide anion (N_3^-).

In S_N1 reaction the rate of the reaction depends on only substrate so after adding nucleophile the rate of reaction is unchanged.

But in S_N2, the rate of reaction depends on both substrate & nucleophile. so by adding nucleophile the rate of reaction also increases. with increasing conc. of nucleophile increasing the rate of reaction in S_N2 modes.

2016, 2024(T), 2015

⊛ The rate of solvolysis of Me_3CBr is found 3×10^4 times faster in 50% aq. ethanol than ethanol (normal) - explain?

⇒ Due to the effect of solvent. H_2O (aqueous) is more polar than ethanol & dielectric constant of H_2O is higher than ethanol so ~~H2O~~ the carbocation will be stable. For this reason rate of reaction is higher than ethanol alone.

$$(CH_3)_3CBr \xrightarrow[-Br^-]{\text{slow}} CH_3-\overset{CH_3}{\underset{CH_3}{\overset{|}{\underset{|}{C}}}}\oplus + Br^-$$

(carbocation)

water molecule can solvate carbocation stabilized by H-bonding. This stabilization lowers the energy barrier for carbocation formation, making it easier for the reaction to proceed. Increased polarity of the solvent mixture enhances the ionization of the halide

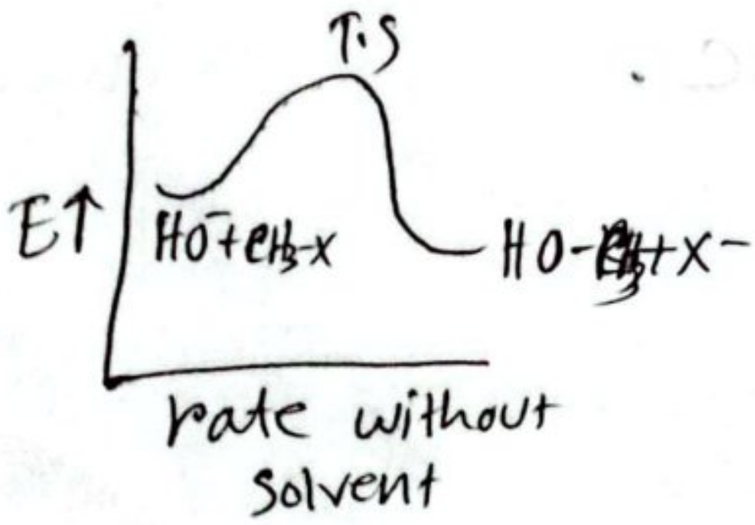
ion, facilitating the formation of the carbocation.

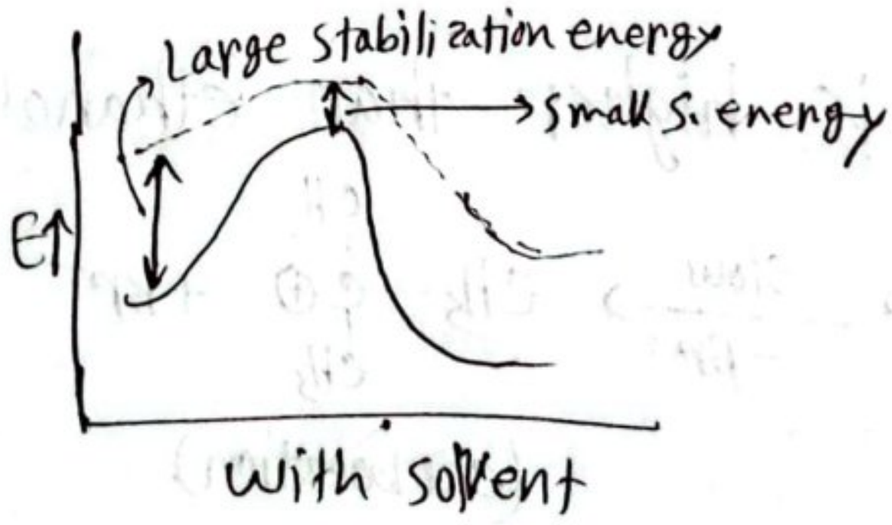
⊗ 2016, 2015 For S_N2, increasing solvent polarity is found to result in Slight decrease in r. rate - why?

⇒ Because in S_N2, new charge is not developed in the transition state & the charge is dispersed in the Transition State comparing the starting material

$$CH_3-X + \bar{O}H \rightarrow \left[HO^{\delta-}\cdots CH_3\cdots X^{\delta-}\right]^{\neq} \rightarrow HO-CH_3 + X^-$$

Polar Solvent solvated starting material stabilized because concentrated charge present but in transition state the charge is dispersed. when solvent polarity increases the more stronger H-bond with form starting material that means stronger solvation is occur in starting material.





⊗ 2016, 2015 why does polar non hydroxylic DMF have marked effect on (increased 4.5×10^9) S_N2 reaction of MeI with N_3^- at 0°C compared the reaction with MeOH?

⇒ The S_N2 reaction of MeI with N_3^- :

$$MeI + N_3^- \xrightarrow[0°C]{S_N2} MeN_3 + I^-$$

For S_N2, the greater the polarity of solvents, the slower the reaction. Again the changing form of protic to aprotic solvents also affect the r. rate. In MeOH, the nucleophile N_3^- is highly solvated through H-bonding; whereas it is very much poorly solvated in DMF. The largely unsolvated N_3^- ion in DMF is a very much stronger nucleophile than the solvated ones which slowers the r. rate. Hence N_3^- in DMF can easily attack MeI & make the reaction 4×10^9 times faster comparing the reaction in MeOH.

2016, 2015, 2021, TU

✱ Why does S_N2 r. rate decreases as the following series is traversed : CH_3-Br, $CH_2(Me)Br$, Me_2CHBr, Me_3C-Br ?

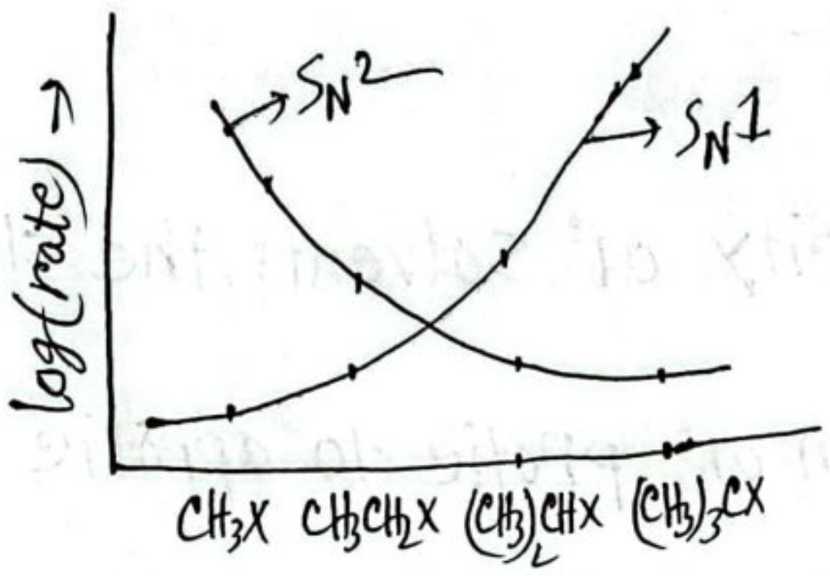
⇒ rate of reaction order :

$$CH_3-Br > MeCH_2Br > Me_2CHBr > Me_3C-Br$$

① Nucleophile attack on the C^{st} carbon & transition state is formed. But more methy group decreas partial positive charge on this carbon, so nucleophile can not easily attack on this carbon.

② more methyl group make more crowding or more steric

~~hin~~ hinderence so the r. rate decreased.



2016

⊗ The π-orbital system of benzene can act as a neibouring group in nucleophilic substitution?

⇒ π-orbital system of benzene can act as a neibouring group by stabilizing intermediates or transition states through deloclization. This occurs when benzene ring is adjacent to the reactive center. If the carbocation is adjacent to a benzene ring, the π-e^- system of the ring can delocalize the +ve charge through resonance. This stabilization makes the intermediate more favourable, increasing the r. rate. The benzene ring's π-e^- can also influence the orientation & approach of the nucleophile due to electronic & steric effects. Ex. solvolysis of benzyl chloride in water. The π-e^- of the benzene ring can participate in the reaction, stabilizing the carbocation intermediate & leading to a faster r. rate.

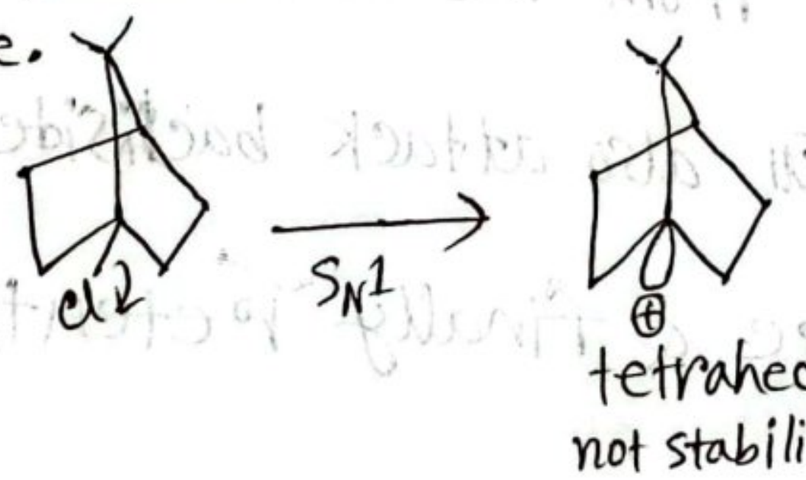
2015, 10

⊛ The following bicyclic halogenoalkane is almost unreactive to S_N1 & S_N2 – why?



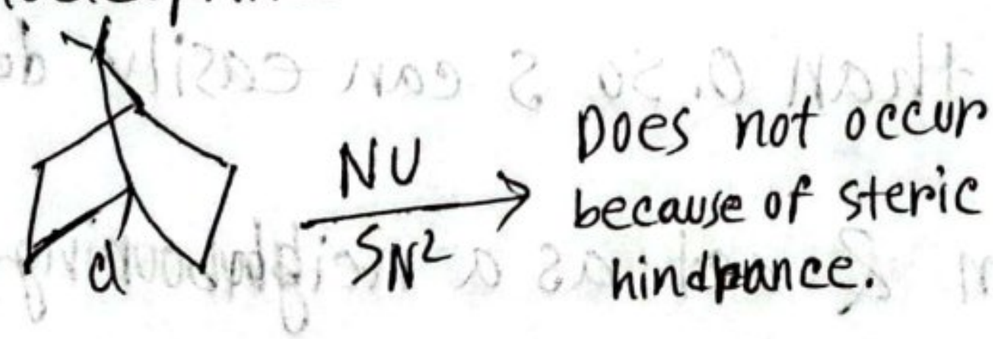
⇒ due to their unique structural constraints, which hinder both the formation of intermediates & the approach of nucleophiles.

Bicyclic structures, particularly those with bridgehead carbons, do not allow for the formation of a stable carbocation due to angle strain (Bredt's rule). normally carbocation planner in geometry but ~~bicy~~ bicyclo cannot for planner structure. so it is unstable & S_N1 mode is not possible. carbocation normary sp^2 but bridgehead 'C' cannot sp^2 due to strain increase.



$$\text{Strain} \propto \frac{1}{\text{stability}}$$

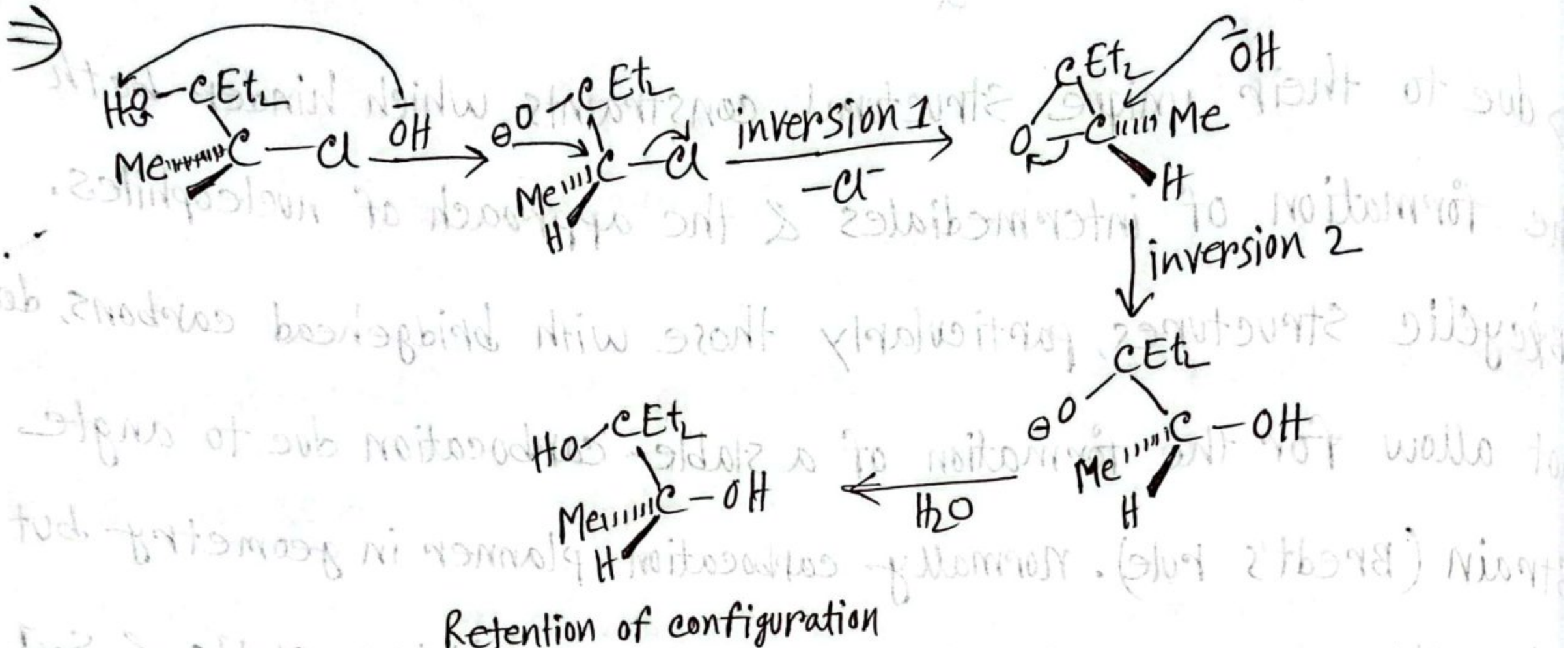
The crowded & rigid bicyclic structure blocks the backside attack of the nucleophile. so S_N2 is very difficult or impossible:



⊛ 2015
In S_N2 order of reactivity $CH_3X > 1° > 2° > 3°$ – why?

⇒ because of steric hindrance.

⊗ 2015
The alkaline hydrolysis of 1,2-chlorohydrin ($MeCHClCEt_2OH$) is found to yield 1,2-diol ($MeCH(OH)CEt_2OH$) with retention - explain?

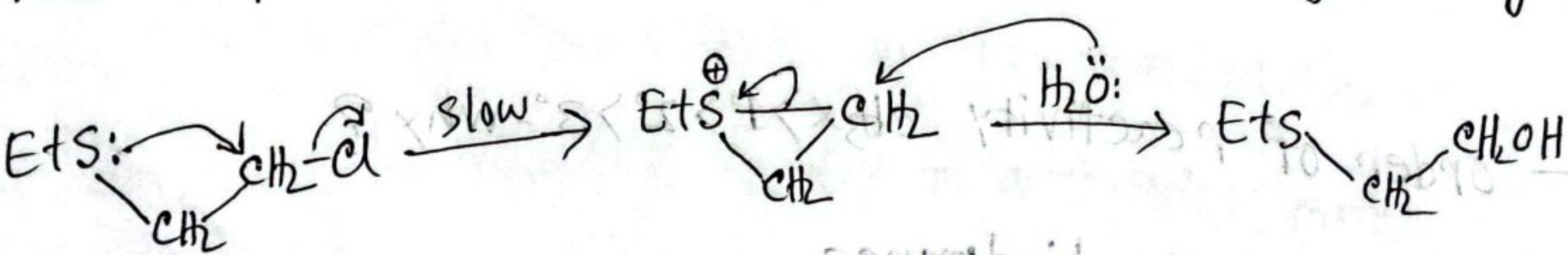


Retention of configuration

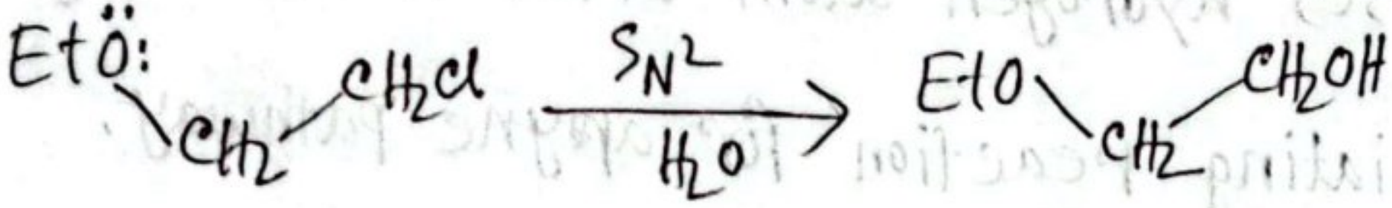
At first $\bar{O}H$ base take H^+ from $HO-CEt_2-$ & O^- is formed. which is neibouring ion & attack from the backside & first inversion take place. The nucleophile $\bar{O}H$ also attack backside of c-o bond & 2nd inversion also take place & finally retention of conf. found.

⊗ 2015, 2017 (+)
$EtSCH_2CH_2Cl$ is found to undergo hydrolysis 10^4 times faster than $EtOCH_2CH_2Cl$ under compareable condition – explain-why?

⇒ S is less electronegative than O. so S can easily donate the e⁻ pair to carbon atom & act as a neighbouring group.

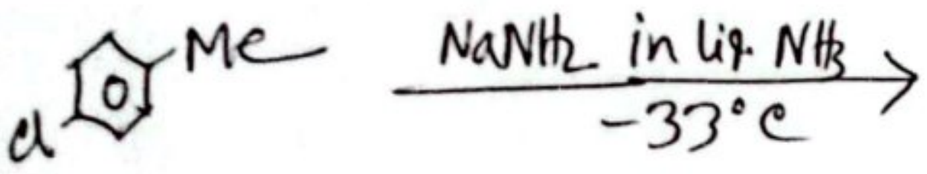


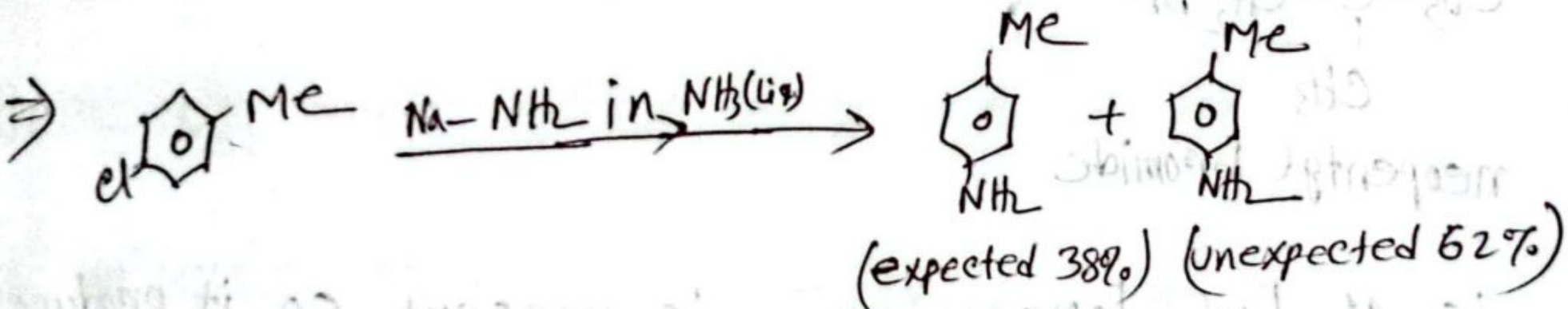
But in $EtOCH_2CH_2Cl$, O (3.5) is sufficiently electronegative & do not want to donate e^- pair & hydrolysis of $EtOCH_2CH_2Cl$ follow S_N2.



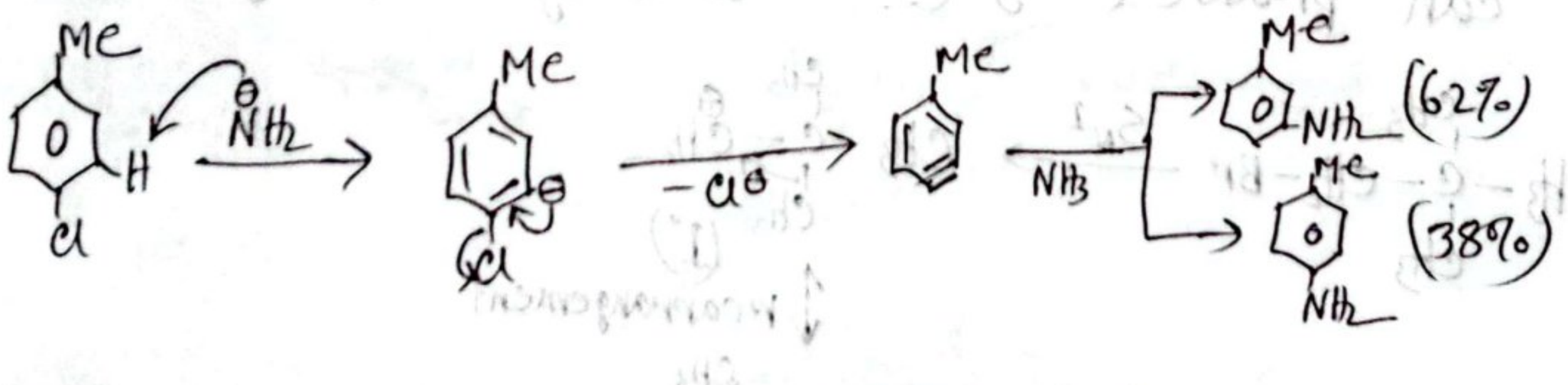
for this difference in electronegativity hydrolysis of $EtSCH_2CH_2Cl$ is 10^4 times faster than $EtOCH_2CH_2Cl$.

2018
(4) complete the reaction & mechanism with evidence:

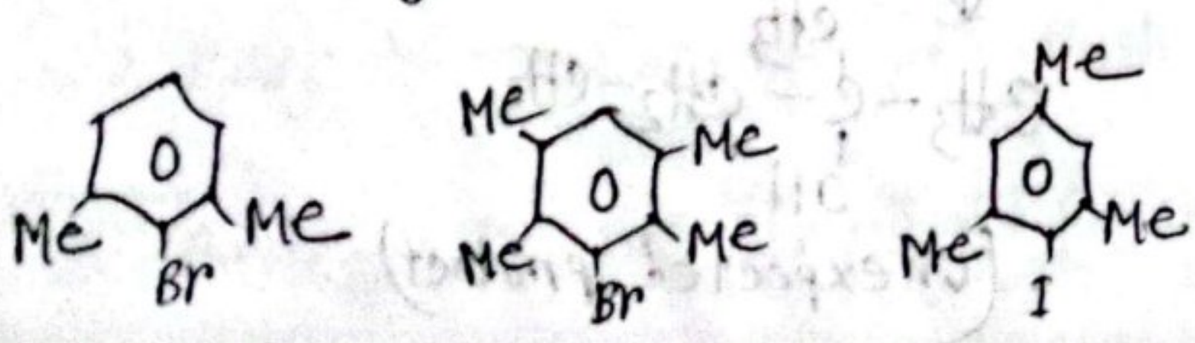




Mechanism:



Evidence: Direct evidence in support of the aryne pathway is provided by the fact that the following halides:-

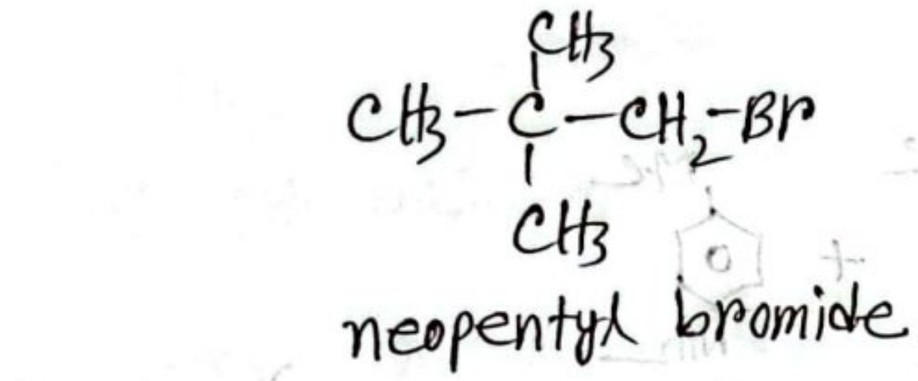


react with $\overset{\ominus}{N}H_2$ only under conditions much more vigorous than can be explained by any steric effect exerted by their o-Me group. Because none possesses hydrogen atom ortho to the halogen which is essential for initiating reaction for aryne pathway.

2018

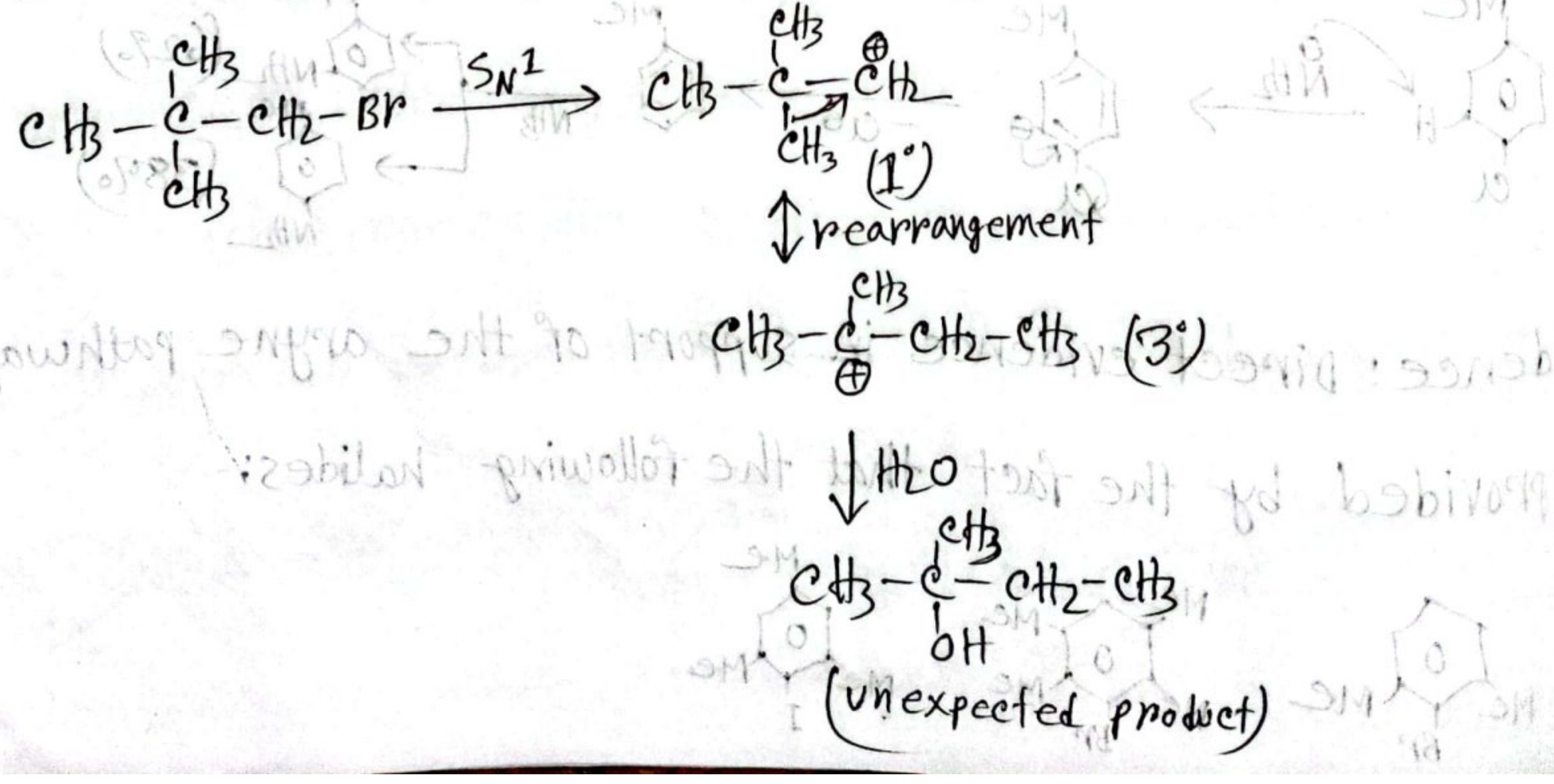
⊛ The S_N2 hydrolysis of neopentyl bromide is slow, but under S_N1 mode the rate is increased & found unexpected product, 2-methyl-2-butanol – explain?

⇒



neopentyl bromide

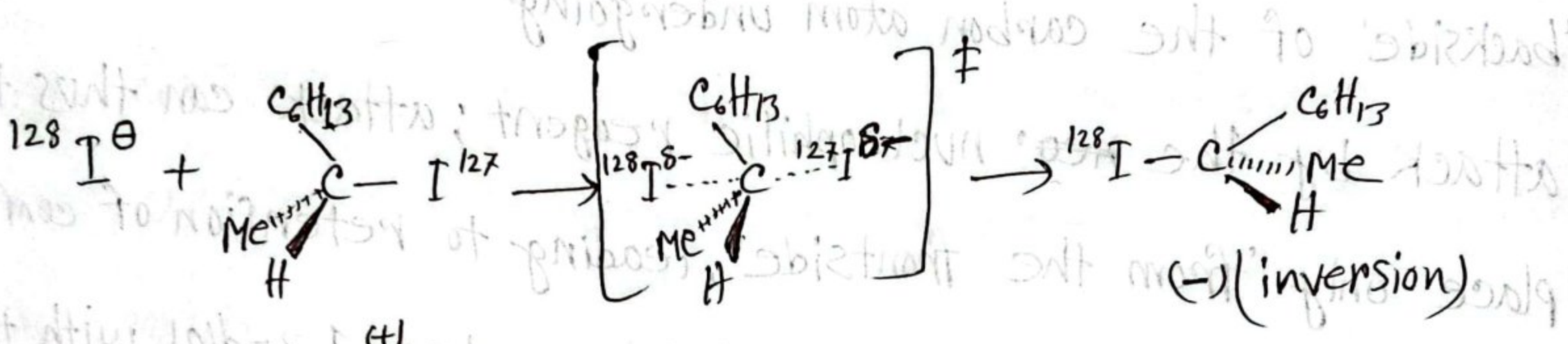
This halide is 1°, but larger group is present so it produce larger steric effect. ~~because~~ so S_N2 becomes slow. But it can produce 3° carbocation by rearrangement,



3° more stable than 1° so rearrangement occur & unexpected 2-methyl-2-butanol is formed.

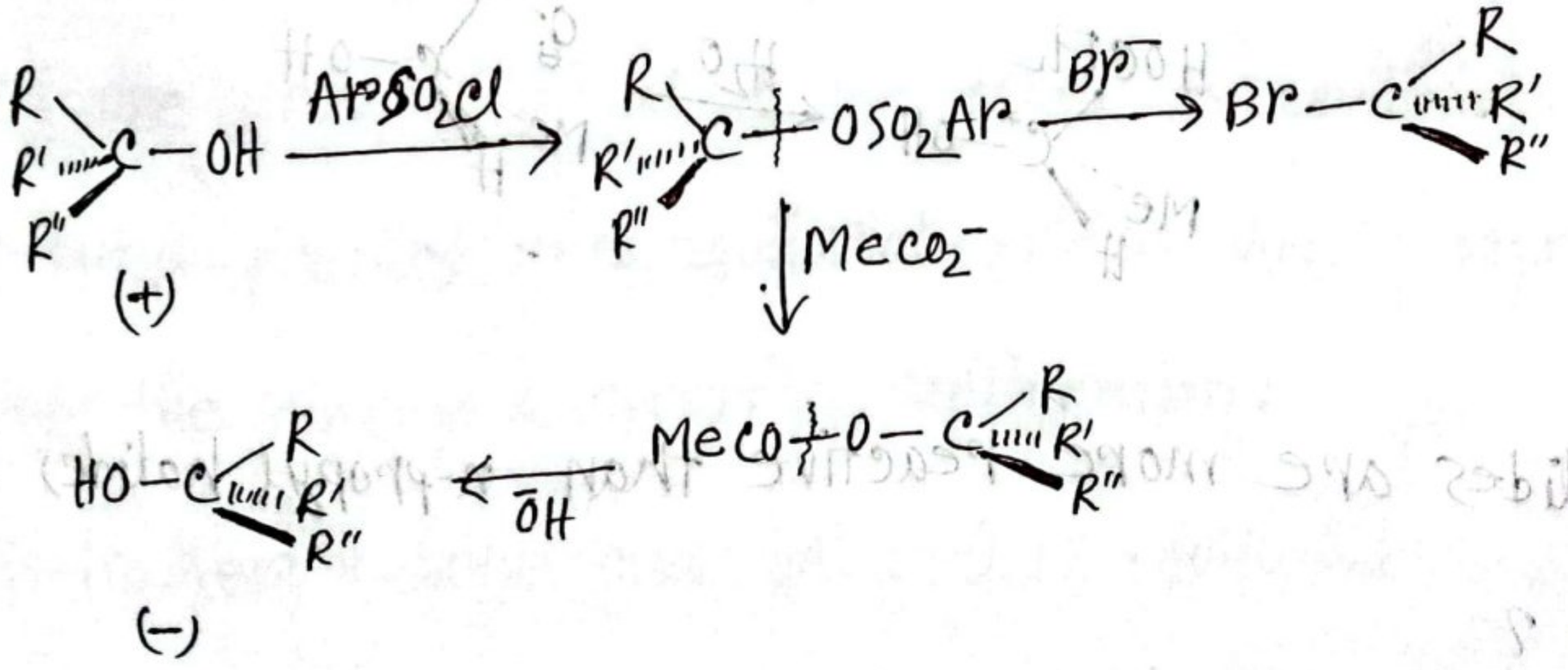
(*) 2018 How can you prove S_N2 involve inversion configuration?

⇒ evidence-1:



Here optically active alkyl halide reacts with labelled $^{128}I^-$ & after reaction opical activity is zero means racemization occur. So S_N2 involve inversion.

evidence-2:



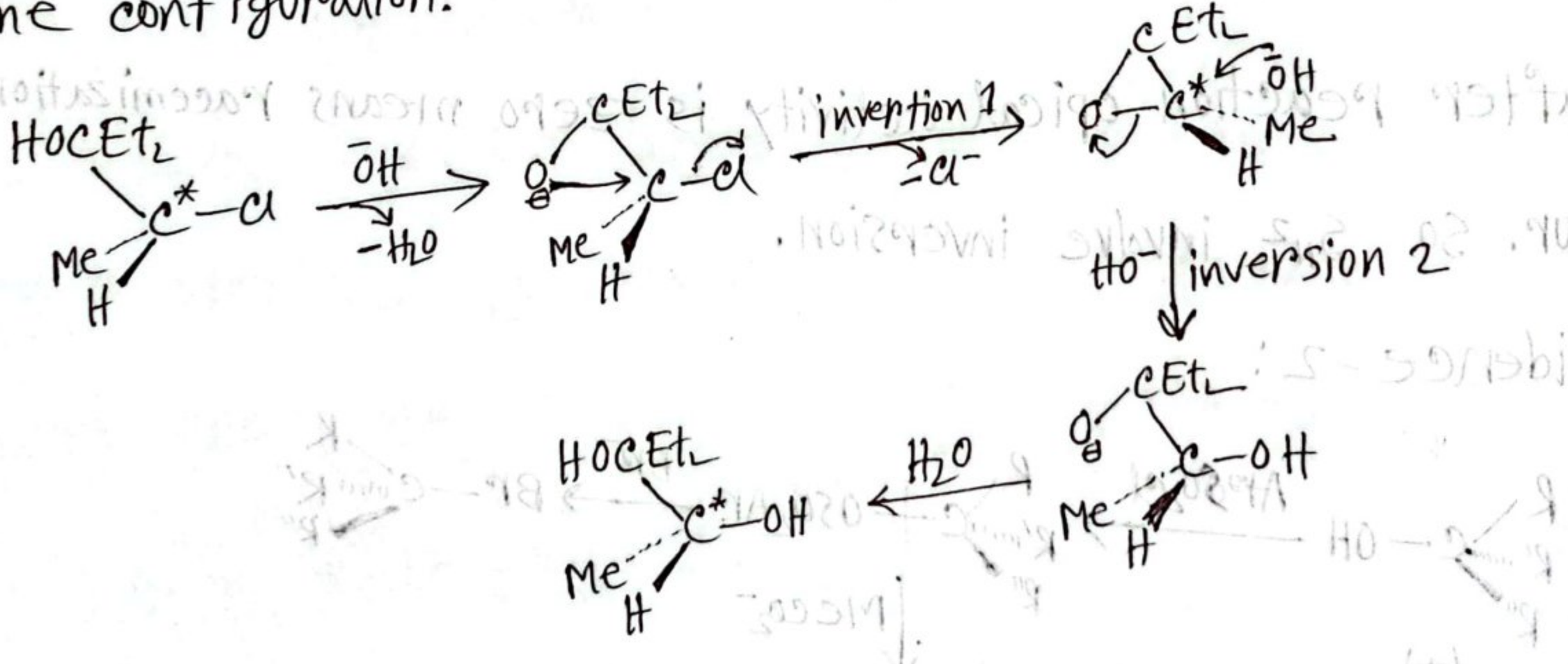
In this reaction (+) Alc react with tosyl chloride ($ArSO_2Cl$) & C–O bond breaking is not involved (C–O bond of Alc is not broken). Their

configuration also same. Then $MeCl^-$ nucleophile attack from th backside in which C-O bond is breaking of nucleophile & inversi occurs & $ArSO_3^-$ expelled.

2018
(*) In Substitution, NGP leads to retension of configuration - explain

⇒ Neibouring group can use its e- pair to interact with the 'backside' of the carbon atom undergoing substitution, thus preventing attack by the ~~neo~~ nucleophilic reagent; attack can thus take place only 'from the frontside', leading to retension of configurat

Ex. base hydrolysis of 1,2-chlorohydrin produce 1,2-diol with the same configuration.



2018
(*) Allyl halides are more reactive than n-propyl halides in S_N1 ? - Why ?

⇒ Due to stability of carbocation. Allylic carbocation is more stabilized by resonance. But n-propyl halide for primary halide

& it has no resonance stability.

$CH_2=CH-CH_2X \xrightarrow{\text{slow}} CH_2=CH-\overset{\oplus}{C}H_2 + X^-$

allyl halide

$CH_2=CH-\overset{\oplus}{C}H_2 \leftrightarrow \overset{\oplus}{C}H_2-CH=CH_2$

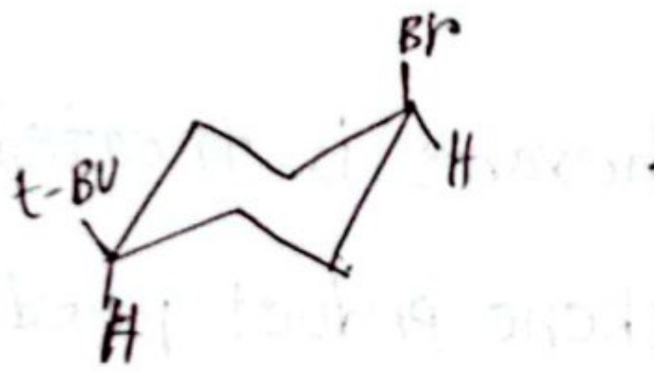
RAJIB
che-50

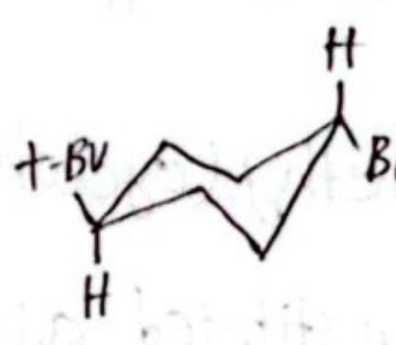
$CH_3-CH_2-CH_2-Cl \xrightarrow{\text{slow}} CH_3-CH_2-\overset{\oplus}{C}H_2 + Cl^-$

(no resonance)

2015, TU, 2020

⊛ Which diasteriomer of 1-bromo-4-t-butylcyclohexane, the cis or trans, undergoes elimination more rapidly when treated with sodium epoxide?



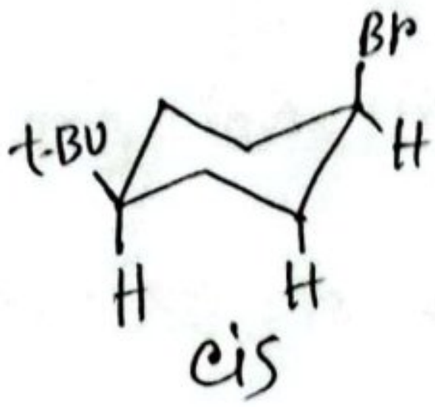


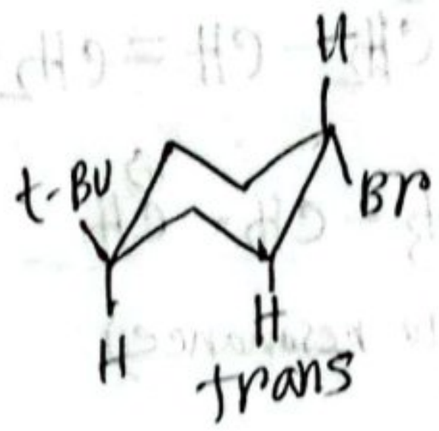
?

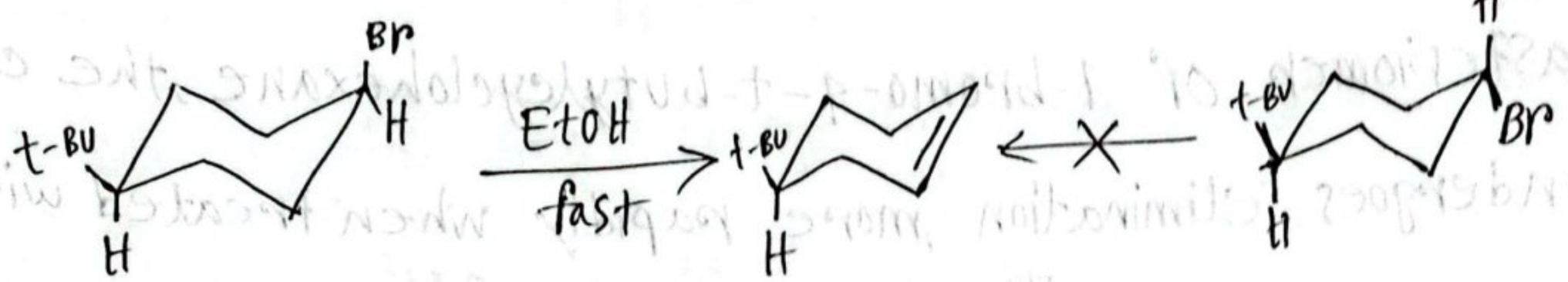
⇒ cis isomer undergo ~~no~~ elimination more rapidly, E_2 requires the β-H to be anti-periplanar to the leaving group (180° apart in the newman projection). Antiperiplanar is easier to achieve when one group is axial & the other is equitorial. The bulky tert-butyl prefers the equitorial position due to steric hindrance, locking the ring in a specific conformation.

In cis-isomer, t-butyl group (at C-4) is equitorial & Br (at C-1) must be axial. this arrangement places at least on β-Hydrogen on the adjacent carbon antiperiplanar to the axial Br, allowing elimination

rapidly. In trans, t-butyl (at c-1) is equitorial, & Br (at c-1) is also equitorial. This arrangement prevents the β-Hydrogen from being anti-periplanar to the Br, making elimination more difficult.

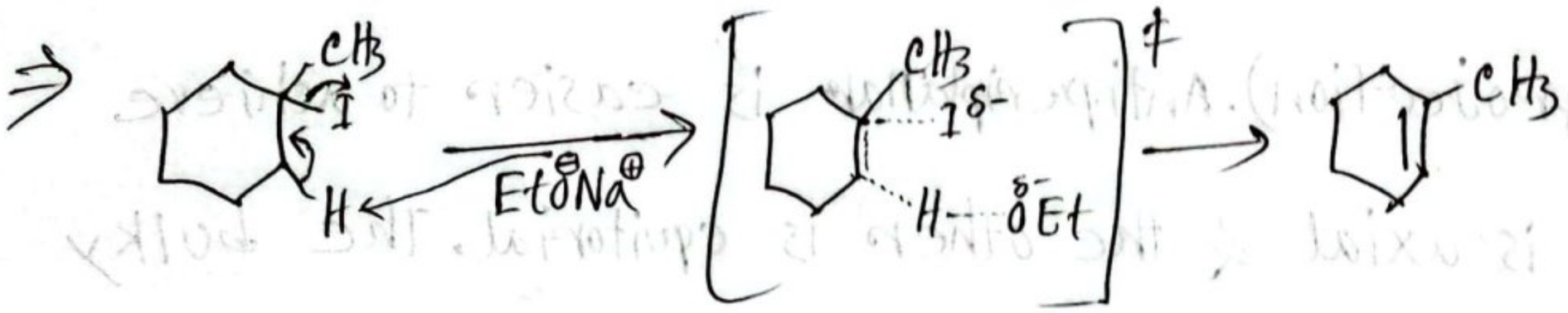




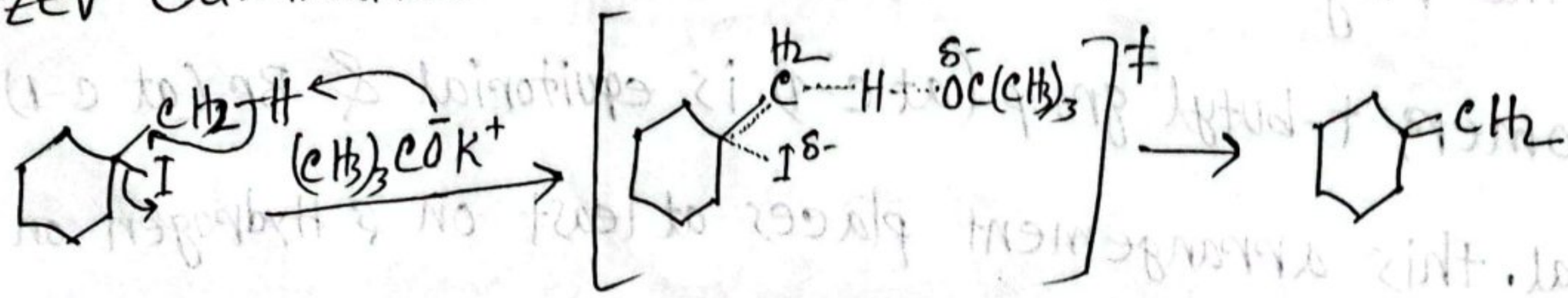


2015, TU, 2020

(✗) When 1-iodo-1-methylcyclohexane is treated with sodium ethoxid the more highly substituted alkene product predominates whereas th same reaction with potassium tert-butoxide gives less substituted alkene predominantly - Give structure of two product & explain?



highly substituted product formed because this reaction follow Saytzev elimination.



Less substituted alkene is formed becase this reaction follow

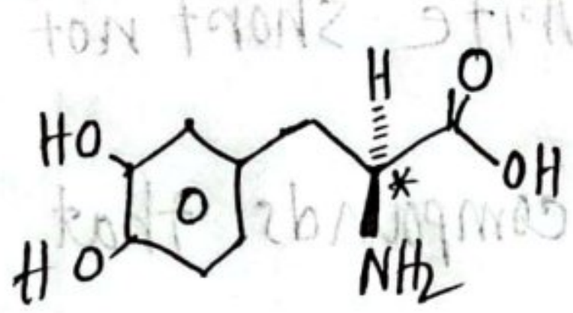
Hoffmann rule. Because this base is larger & present steric factor so it absorbs proton from $-CH_3$ group & less substituted product (alkene) is formed.

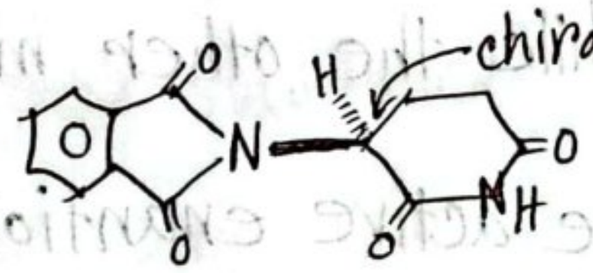
(✱) 2021 "Importance of ~~chemical~~ chiral drugs" write short not about it ?

⇒ chiral drugs are pharmaceutical compounds that exist in two mirror-image forms, known as enantiomers. The importance of ~~two~~ chiral drugs lies in their ability to enhance therapeutic outcomes & reduce side effects. one enantiomer may provide the desired therapeutic effect, while the other might be less effective. By isolating & using the active enantiomer, drug efficiency can be improved & patient safety ensured.

Drugs having chiral centers are called chiral drugs. ⓘ chirality is a crucial factor which can impact drug's safety, efficacy, pharmacological activity & metabolism. ⓘⓘ chirality affects a drug's interactions with biological targets, such as proteins, nucleic acids. ⓘⓘⓘ chiral drugs have applications in treating a variety of conditions including cardiovascular disease, CNS disorders, local anesthesia, anti-~~inf~~ inflammatories, antimicrobials & anticancer drugs. ⓘⓥ chiral drug Simplifies the interpretation of the Basic pharmacology ⓥ our body-

contains large number of chiral molecule so chiral drug can interact with them properly. (VI) By using only the active enantiomer, lower dosage may achieve the desired effect, reducing side effects & improving patient outcomes.

EX. L-Dopa:  which is used to treat parkinson's disease (L-enantiomer), causes decreasing dopamine production. Brain converts L-Dopa to dopamine, associated with movement & emotions. while D-enantiomer is inactive.

Thalidomide :  R-enantiomer acts as a sedative, while the S-enantiomer caused severe birth defects.

2018
(★) chemical shift value of Aldehydic proton is found 9 to 10- why?

⇒ (I) Aldehydic H+ directly attached to a carbon that is part of carbonyl group. 'O' is highly electronegative, pulling e- density away from the aldehydic proton. This results in significant deshielding of the proton, shifting its chemical shift downfield (higher PPM)

(II) π-e cloud of the carbonyl group creates an anisotropic magnetic field. This field deshields Ald. proton, contributing to its downfield shift. This increase the deshielding effects,

making signal in 9 to 10 ppm ranges.

2017
(**) Differentiate btwn transition state & intermediate with example?

→

Transition State	intermediate
It has partial bonds	It has complete bonds.
It has higher energy than intermediate	It has lower energy than T.S.
It has no definite geometry	It has definite geometry.
can not be isolated	can be isolated
Extremely unstable	Relatively stable
Represented by a double dagger symbol (‡).	Represented as a regular molecular structure

CamScanner